Exercise – 9: Implement a Case on Windows and Linux Forensics

Aim

To perform a digital forensic investigation on Windows and Linux systems by collecting and analyzing digital evidence using forensic tools.

Objectives

- To understand the concept of Digital Forensics.
- To learn how to acquire forensic images of storage devices.
- To analyze files, logs, and user activities from Windows and Linux systems.
- To use forensic tools for evidence investigation.

Software Requirements

- Windows Operating System
- Linux Operating System (Kali Linux / Ubuntu)
- Forensic Tools:
    - Autopsy (Digital Forensics Platform)
    - FTK Imager (Forensic Disk Imaging Tool)
    - Volatility (Memory Forensics Framework)

Hardware Requirements

- Computer system with minimum 8GB RAM
- External storage device for storing forensic images

Theory

Digital Forensics

Digital Forensics is the process of identifying, collecting, preserving, analyzing, and presenting digital evidence from computers or digital devices.

Digital forensics is commonly used in:

- Cybercrime investigations
- Incident response
- Corporate investigations
- Law enforcement cases

During forensic analysis, investigators must ensure:

1. Evidence integrity is maintained
2. Original data is not modified
3. Proper documentation is maintained

Forensic Investigation Process

1. Identification – Identify potential sources of evidence
2. Acquisition – Create forensic images of data
3. Preservation – Maintain integrity of evidence
4. Analysis – Examine data using forensic tools
5. Documentation – Record findings and observations

PART A – Windows Forensics

Tool Used: FTK Imager

About the Tool: FTK Imager is a digital forensic tool used to create forensic images of storage devices and analyze files without altering original evidence.

It supports:

- Disk imaging
- File preview
- Evidence acquisition
- Deleted file recovery

Steps to Use FTK Imager

Step 1: Install FTK Imager

1. Download FTK Imager from official website.
2. Install the software on the Windows system.

Step 2: Create a Disk Image

1. Open FTK Imager.
2. Click File → Create Disk Image.
3. Select Physical Drive or Logical Drive.
4. Select the target drive.
5. Choose Image Type (E01 recommended).
6. Enter case information:
    - Case number
    - Examiner name
    - Evidence description
7. Select destination folder.
8. Start imaging process.

The tool will create a forensic copy of the drive.

Step 3: Analyze the Image

1. Open FTK Imager.
2. Click File → Add Evidence Item.
3. Select the forensic image file.
4. Browse through directories.

Investigators can analyze:

- Deleted files
- Documents
- Images
- Registry entries
- Browser history

Step 4: Recover Deleted Files

1. Navigate through folders.
2. Right-click deleted files.
3. Select Export Files.
4. Save recovered files for analysis.

PART B – Linux Forensics

Linux systems store useful forensic artifacts in log files and command histories.

Step 1: Open Terminal

Launch the terminal in the Linux system.

Step 2: Analyze File System

List files and directories:

ls -la

This command shows:

- Hidden files
- File permissions
- File ownership

Step 3: Check Command History

history

This command displays commands executed by users.

Investigators analyze this to identify suspicious activities.

Step 4: Analyze Authentication Logs

cat /var/log/auth.log

This file contains:

- Login attempts
- Failed login attempts
- SSH login records

Suspicious repeated login failures may indicate a brute-force attack.

Step 5: Check System Logs

cat /var/log/syslog

This log contains system events such as:

- service start/stop
- system errors
- hardware activity

Memory Forensics using Volatility
Volatility is used to analyze RAM memory dumps.
Memory analysis helps investigators detect:

- running processes
- malware in memory
- network connections

Step 1: Obtain Memory Dump
Memory dump can be captured using forensic tools.

Step 2: Analyze Memory Image
Example command:
volatility -f memorydump.mem pslist
This command lists running processes.

Example command:
volatility -f memorydump.mem netscan
This displays active network connections.

Observation

System	Evidence Collected
Windows	Deleted files, browser history, registry artifacts
Linux	Command history, login logs, system logs
Memory	Running processes and network connections

Applications

- Cybercrime investigation
- Incident response analysis
- Malware investigation
- Corporate security audits

Precautions

- Never analyze the original evidence directly.
- Always work on forensic copies.
- Maintain a proper chain of custody for evidence.

Result

- Thus, a forensic investigation case was implemented on Windows and Linux systems. Digital evidence such as system logs, deleted files, and user activities were successfully collected and analyzed using forensic tools.

Viva Questions

1. What is digital forensics?
2. What is forensic imaging?
3. What is the purpose of FTK Imager?
4. What information can be obtained from Linux log files?
5. What is memory forensics?
6. What is the role of Volatility in forensic analysis?